SUMMARY REPORT

Objective

The main objective of this project was to analyze global pollution data and classify countries into different pollution severity levels—Low, Medium, and High. The aim was also to understand how pollution impacts energy consumption and recovery, and to derive meaningful insights for environmental improvement.

we carried out the project in a structured, multi-phase approach:

- **Data Preprocessing**
  - Imported and cleaned the dataset by handling missing values and outliers.
  - Standardized and normalized important features like CO_2 emissions and industrial waste.
  - Encoded categorical variables such as country and year.
  - Created new features like energy consumption per capita and pollution trends.
  - Applied feature scaling to ensure consistency across pollution indices.
- **Model Implementation**
  - Applied three machine learning algorithms:
    - **Naive Bayes** for probabilistic classification.
    - **K-Nearest Neighbors (KNN)** for similarity-based classification.
    - **Decision Tree** for rule-based classification.
  - Performed hyperparameter tuning to improve model performance.
  - Evaluated all models using metrics like Accuracy, Precision, Recall, F1-score, and Confusion Matrix.
- **Analysis and Visualization**
  - Compared model performances to identify the most effective classifier.
  - Visualized results using confusion matrices and classification reports.
  - Analyzed pollution patterns and their relationship with energy recovery.

Outcome

- Successfully classified countries into pollution severity categories using multiple machine learning models.
- Identified the most accurate and efficient model through comparative analysis.
- Gained insights into how pollution levels affect energy consumption and recovery.
- Provided actionable recommendations for reducing pollution and improving sustainability.
- Delivered a complete solution including a Jupyter Notebook, visualizations, and analytical findings.